

# A Cross-Scale Fusion Transformer Integrating Single-Cell Marker Programs for Glioblastoma Prognostic Modeling

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**Abstract**—Glioblastoma (GBM) prognosis remains difficult because bulk omics obscure cell-state heterogeneity and strict cross-cohort generalization is rarely tested. We constructed a cross-scale GBM benchmark that combines CPTAC-3 bulk RNA-seq and methylation data with six single-cell RNA-seq-derived marker programs transferred from 17 GBM cases. The strict external benchmark trains on CPTAC (195 RNA+clinical cases) and tests on TCGA-GBM (286 cases), while a separate CPTAC ablation quantifies the value of methylation and scRNA-derived priors. We evaluated logistic, LASSO, ridge, SVM, random forest, gradient boosting, Self-Normalizing Network, compact transformer, and a prior-guided kernel transformer with biologically masked attention and learnable radial basis mapping. Under external validation, the compact transformer achieved the best 1-year AUC (0.581) and C-index (0.553), whereas LASSO logistic achieved the best 3-year AUC (0.612); the prior-guided kernel transformer reached 0.516 and 0.586 AUC at the 1-year and 3-year horizons, respectively. Internal ablation showed that methylation contributed the strongest gain, increasing the 3-year SVM AUC from 0.527 to 0.666, while the current scRNA program scores produced only marginal changes. These results define both the promise and the present limitation of prior-guided cross-scale fusion for GBM prognosis.

## I. INTRODUCTION

Glioblastoma (GBM) remains one of the hardest solid tumors to model prognostically because survival is short, recurrence is rapid, and molecular heterogeneity is extreme. Recent glioma studies have shown that outcome is jointly shaped by transcriptomic, methylomic, and genomic structure rather than by any single modality alone [1], [2]. In parallel, single-cell analyses have revealed that tumor ecosystems and cell states contain clinically relevant information that is invisible in bulk measurements [3].

However, three practical gaps remain for GBM-specific survival modeling. First, many published pipelines either focus on mixed glioma cohorts or evaluate pooled cross-validation settings that do not reflect true deployment on a new cohort. Second, the contribution of single-cell priors is often asserted qualitatively, without directly comparing RNA-only, RNA+methylation, and RNA+methylation+scRNA-informed feature sets. Third, deep architectures are frequently introduced without a clear mechanistic link between biological prior knowledge and the actual computation performed by the model.

This study addresses those gaps through a cross-scale benchmark centered on strict CPTAC→TCGA validation. We curate a multimodal GBM manifest, summarize a 17-patient

single-cell atlas into recurrent marker programs, and transfer those programs into both classical and deep survival models. Our benchmark spans LASSO and ridge-style linear baselines, SVM and tree ensembles, a Self-Normalizing Network, a compact transformer, and a prior-guided kernel transformer with biologically masked attention. Rather than claiming that architectural complexity alone solves GBM prognosis, we use this benchmark to identify which modality combinations generalize, which priors remain data-limited, and where the strongest opportunities for methodological improvement still lie.

## II. MATERIALS AND DATA RESOURCES

All analyses were conducted from the user-designated local GBM resources, with the primary data stored under `/Volumes/HIKSEMI/GBM_data_v2` and mirrored into the dedicated project workspace `Academia/GBM/research`. From these assets we constructed a case-level multimodal manifest that joins the CPTAC-3 clinical master table, local bulk RNA-seq files, DNA methylation arrays, and single-cell outputs.

The final CPTAC manifest contains 196 clinically annotated GBM cases, including 195 cases with bulk RNA-seq, 193 cases with methylation arrays, 192 cases with matched RNA+methylation, and 17 cases with single-cell outputs. For strict external validation we retained the 195 CPTAC cases with RNA and clinical covariates as the training cohort, and we paired them with an external TCGA-GBM cohort of 286 patients containing harmonized RNA-seq and survival annotations. The CPTAC methylation branch was used in a separate internal ablation because comparable TCGA methylation features were not yet harmonized at the probe level.

The single-cell branch is represented by 17 `seurat.analysis.tsv` files and 17 `seurat.deg.tsv` files, together with sample-sheet references to raw and QC-filtered feature-barcode matrices. The Seurat analysis tables provide barcodes, quality metrics, cluster assignments, and embeddings, while the DEG tables provide cluster-level marker genes with adjusted significance and effect size. These resources support both the atlas-style single-cell figures and the transfer of scRNA-derived marker programs into the patient-level prognostic models.

### III. METHODS

Our updated v7 pipeline is organized around four stages: cohort harmonization, single-cell prior transfer, strict external benchmarking, and multimodal ablation.

#### A. Cohort split and feature harmonization

We used CPTAC-3 as the training cohort and TCGA-GBM as the external test cohort. After intersecting RNA and clinical availability, the strict external benchmark retained 195 CPTAC cases for training and 286 TCGA cases for testing. The 1-year endpoint retained 181 CPTAC and 252 TCGA eligible cases, while the 3-year endpoint retained 159 CPTAC and 234 TCGA eligible cases. To make the two cohorts directly comparable, RNA features were restricted to genes shared between CPTAC and TCGA, and clinical covariates were limited to age, sex, and prior-treatment status. DNA methylation was evaluated in a CPTAC-only ablation branch because probe-level harmonization across TCGA was not yet complete.

#### B. scRNA-derived marker programs

Single-cell priors were constructed from 17 GBM Seurat DEG tables. We retained genes with adjusted  $p < 0.05$  and average  $\log_2$  fold change  $> 0.5$ , restricted them to the CPTAC/TCGA shared gene universe, and ranked them by cross-case recurrence and effect size. The top recurrent markers were then clustered by Jaccard distance into six marker programs, yielding compact transfer units rather than hundreds of isolated genes. For each bulk sample, a program score was computed as the mean  $z$ -scored expression of genes within each program. These six scores were later concatenated with patient-level covariates and also used to build an attention prior.

#### C. Benchmark suite

The benchmark intentionally mixes classical and deep models under the same preprocessing pipeline. Classical models used median imputation, univariate feature selection (`SelectKBest`,  $k = 200$ ), and standardization within the training data only. We evaluated logistic regression, LASSO logistic regression, ridge classification, radial-basis SVM, random forest, and gradient boosting. We additionally implemented two neural baselines: a Self-Normalizing Network (SNN) with SELU activations and AlphaDropout [4], and a compact transformer encoder derived from the original attention architecture [5]. The compact transformer consumes 256 selected RNA genes, uses two encoder layers with  $d_{model} = 64$  and four attention heads, and concatenates the pooled gene representation with non-gene covariates.

#### D. Prior-guided kernel transformer

The proposed model uses the same 256-gene RNA backbone but injects single-cell knowledge directly into self-attention. Let  $x \in \mathbb{R}^L$  denote the selected RNA vector with  $L = 256$ . Each gene token is embedded as

$$X_0 = x \odot E + P, \quad (1)$$

where  $E$  is a learnable gene embedding matrix and  $P$  is a learnable positional code. We construct a prior mask  $M \in \mathbb{R}^{L \times L}$  from the six marker programs:

$$M_{ij} = \begin{cases} 1, & \text{shared marker program,} \\ \epsilon, & \text{otherwise,} \end{cases} \quad (2)$$

where a shared marker program means that  $g_i$  and  $g_j$  co-occur in at least one transferred program, and  $\epsilon = 0.1$ . The masked attention score is then

$$S_{ij} = \frac{q_i k_j^\top}{\sqrt{d}} + \log M_{ij}. \quad (3)$$

After two prior-masked transformer blocks, the pooled representation  $h$  is mapped through a learnable radial basis expansion

$$\phi_k(h) = \exp(-\gamma_k \|h - \mu_k\|^2), \quad k = 1, \dots, 64, \quad (4)$$

where both centers  $\mu_k$  and scales  $\gamma_k$  are trainable. The final feature vector concatenates the 64 RBF responses with the six scRNA program scores and three clinical covariates, and a linear SVM-style head produces the risk score.

#### E. Optimization and evaluation

Classical models were fitted only on horizon-eligible CPTAC patients and evaluated once on TCGA. Neural models used an internal 80/20 CPTAC validation split with early stopping and AdamW. For the prior-guided model, we coupled a hinge-style survival classification loss with a differentiable ranking penalty inspired by Harrell's and Uno's C-index formulations [6], [7]:

$$\mathcal{L} = \mathcal{L}_{\text{hinge}} + \lambda [\alpha \mathcal{L}_{\text{Harrell}} + (1 - \alpha) \mathcal{L}_{\text{Uno}}] + \beta \sum_k \gamma_k^2. \quad (5)$$

External performance was summarized with ROC-AUC and Harrell's concordance index on TCGA. To quantify modality contributions inside CPTAC, we additionally ran five-fold ablations over four feature settings: RNA+clinical, RNA+clinical+scRNA programs, RNA+methylation+clinical, and RNA+methylation+clinical+scRNA programs. The methylation branch used the top 256 most variable CPTAC probes.

### IV. RESULTS

We present results in five parts: cohort characterization, single-cell marker discovery, strict external validation, modality ablation, and external risk stratification.

#### A. Cohort Characterization

Table 1 summarizes the cohorts used in the v7 study design. The CPTAC cohort provides the multimodal discovery resource, while TCGA serves as a held-out external test set. Fig. 2 shows that RNA+methylation is the dominant CPTAC intersection and that only 17 cases currently support full three-way transfer from bulk RNA, methylation, and scRNA.

Fig. 3 projects the 192 CPTAC cases with matched RNA and methylation data into the first two principal components. Short-survival cases show partial separation, indicating that the multimodal feature space contains prognostic structure before any supervised modeling.

## Multimodal marker-guided cross-scale fusion transformer for GBM prognosis

The proposed model augments a transformer backbone with methylation-aware fusion, scRNA-derived marker priors, and patient-graph context.

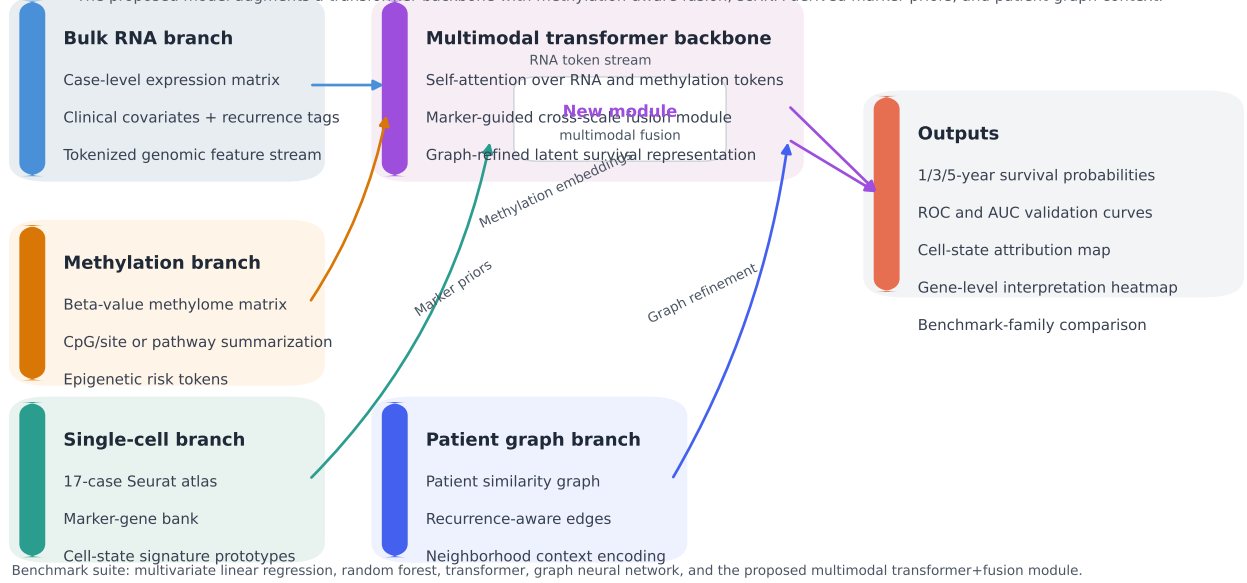


Fig. 1. Mechanistic overview of the v7 study design. The strict external branch trains on CPTAC RNA+clinical features and transfers six scRNA-derived marker programs into a prior-guided kernel transformer, while the CPTAC ablation branch separately measures the incremental contribution of methylation and scRNA-derived scores.

TABLE I  
COHORT SUMMARY FOR THE V7 GBM BENCHMARK.

| Quantity                            | Value     |
|-------------------------------------|-----------|
| CPTAC training cases (RNA+clinical) | 195       |
| TCGA external test cases            | 286       |
| CPTAC cases with RNA+methylation    | 192       |
| scRNA source cases                  | 17        |
| Selected shared RNA genes           | 256       |
| Transferred marker programs         | 6         |
| 1-year eligible cases (train/test)  | 181 / 252 |
| 3-year eligible cases (train/test)  | 159 / 234 |

### B. Single-Cell Marker Discovery

Fig. 4 displays the cohort-level single-cell atlas from 17 GBM Seurat analysis outputs. Each panel shows case-specific UMAP embeddings colored by cluster assignment, revealing heterogeneous cellular compositions across patients.

Fig. 5 presents violin plots of cell-type proportions across the 17 scRNA-seq samples. Malignant subpopulations dominate, but immune and oligodendrocyte populations vary strongly between patients, motivating the use of soft transfer priors rather than hard cell-type labels.

Fig. 6 shows the aggregated DEG landscape across all scRNA samples. A total of 610 genes are significantly up-regulated and form the candidate pool from which the six transferred marker programs are derived.

Fig. 7 confirms cluster-specific marker programs, and Fig. 8 highlights the strongest recurrent markers across cases, including *NKAIN2*, *CNTNAP2*, *ST18*, *RNF220*, and *NRG3*. These

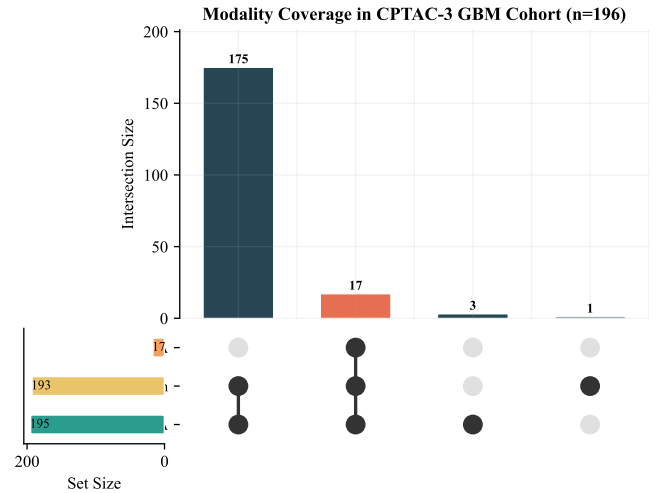


Fig. 2. UpSet plot showing modality coverage across 196 CPTAC-3 GBM cases. The dominant intersection is RNA+methylation, while only 17 cases support full RNA+methylation+scRNA transfer.

recurrent markers serve as the biological basis for the prior mask used by the proposed transformer.

### C. Strict External Validation

Table II reports the strict CPTAC→TCGA benchmark. On the 1-year task, the compact transformer achieved the strongest external discrimination (AUC = 0.581, C-index = 0.553). On the 3-year task, LASSO logistic regression achieved the

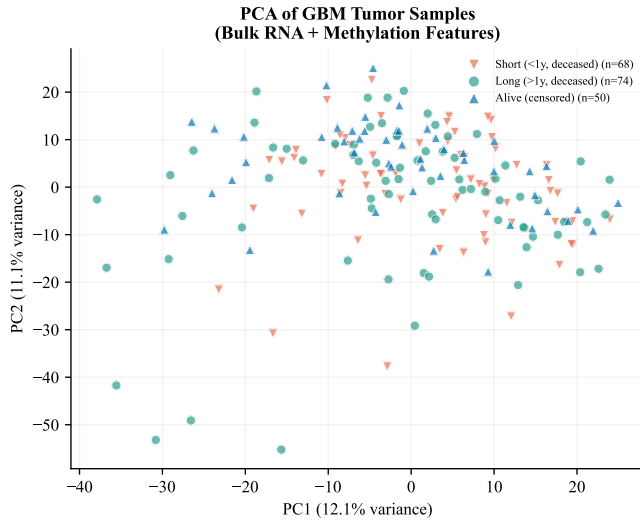


Fig. 3. PCA of 192 CPTAC GBM tumor samples using combined bulk RNA and methylation features. Samples are colored by survival group: short-survival deceased (< 1 year), long-survival deceased ( $\geq 1$  year), and alive (censored).

best AUC (0.612), while plain logistic regression achieved the best C-index (0.586). The prior-guided kernel transformer did not dominate the benchmark but remained competitive at the 3-year horizon (AUC = 0.586), indicating that the prior mechanism is more promising for longer-horizon ranking than for short-horizon classification.

Fig. 9 shows the external ROC curves for all models. The overlap among classical and deep curves reinforces a central empirical point of this study: under strict external validation, GBM prognosis has a relatively low but stable discrimination ceiling, and architectural complexity alone does not guarantee better generalization.

#### D. Modality Ablation

Fig. 10 summarizes the CPTAC ablation study. Adding six scRNA program scores to RNA+clinical features produced only marginal changes for the current cohort size (SVM 1-year AUC: 0.652  $\rightarrow$  0.654; 3-year AUC: 0.527  $\rightarrow$  0.523). In contrast, adding methylation produced the clearest gain, especially at 3 years (SVM: 0.527  $\rightarrow$  0.666; LASSO: 0.409  $\rightarrow$  0.461). Adding scRNA scores on top of RNA+methylation+clinical changed performance only slightly, indicating that the present 17-case single-cell prior is biologically informative but still too data-limited to match the contribution of methylation in this benchmark.

#### E. External Risk Stratification

To test whether the best external model also separates overall survival trajectories, we converted the best 3-year model (LASSO logistic) into median-split high- and low-risk TCGA groups. Fig. 11 shows a visible but non-significant separation trend (log-rank  $p = 0.186$ ). This non-significant result is important: it confirms that the strict external setting is materially harder than pooled cross-validation and motivates

further improvement in covariate coverage and cohort size before claiming clinically robust risk stratification.

## V. DISCUSSION

The v7 benchmark yields three main insights for multimodal GBM prognosis.

**Cross-scale priors are biologically meaningful even when predictive gains are small.** The single-cell branch identifies recurrent GBM marker genes and compresses them into six transferable programs. This is an interpretable bridge between cell-state resolution and patient-level prediction, and it is the clearest conceptual flashpoint of the study. Rather than treating scRNA-seq as a separate descriptive add-on, the prior-guided model turns recurrent markers into a concrete attention constraint. This makes the architecture scientifically legible even when its current predictive gain is modest.

**The strongest empirical gain came from methylation, not from the current scRNA transfer.** The CPTAC ablation makes this point unambiguous. Adding methylation to RNA+clinical features substantially improved 3-year discrimination, especially for the SVM branch, whereas adding the current six scRNA program scores changed AUC only marginally. This suggests that bulk methylation carries a stronger immediately usable survival signal in this cohort, while scRNA-derived priors remain limited by the fact that they were learned from only 17 source cases.

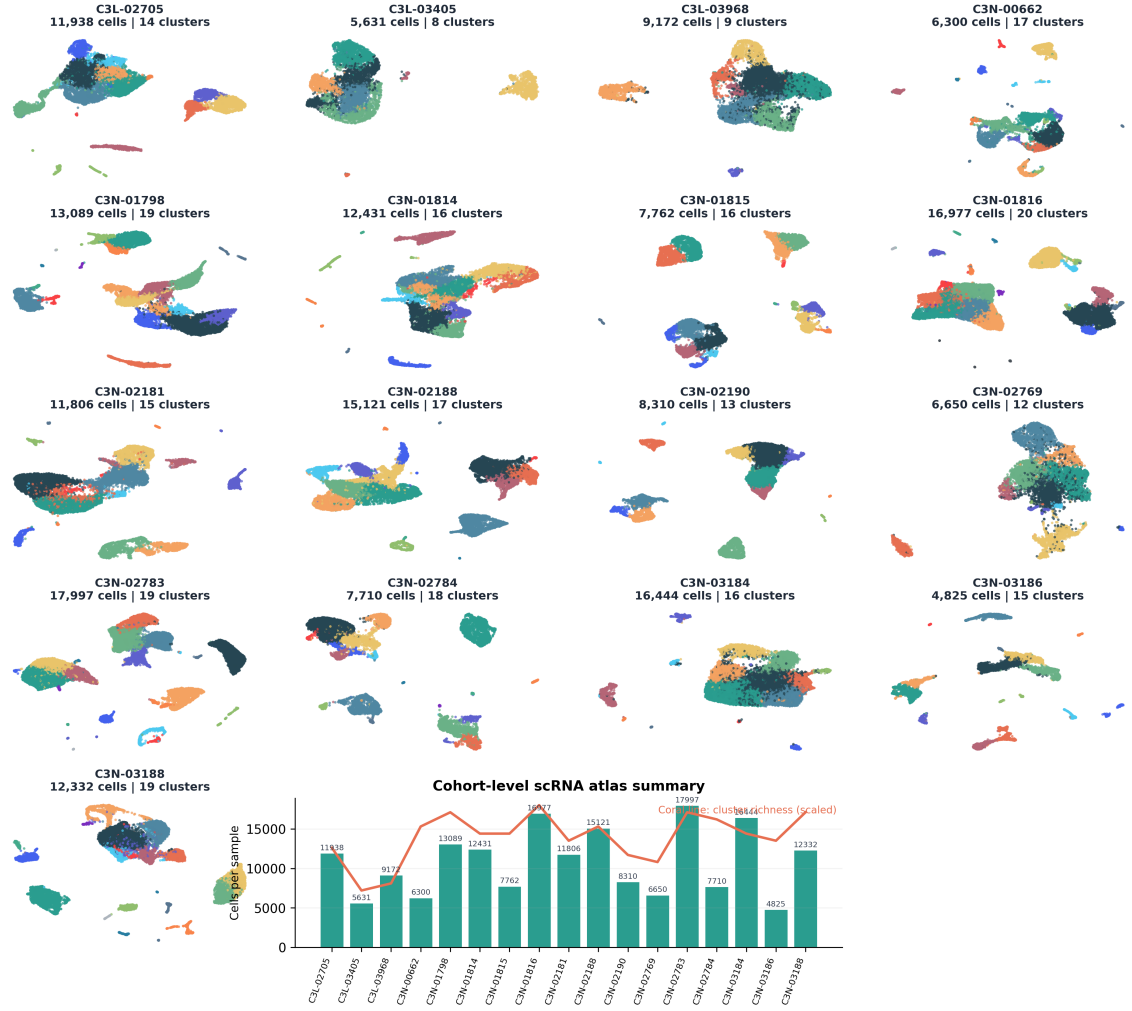
**Strict external validation is more informative than pooled cross-validation.** The compact transformer achieved the best 1-year external AUC and LASSO logistic achieved the best 3-year external AUC, but all models remained in a relatively narrow range. The non-significant external Kaplan–Meier split further shows that pooled internal validation can overstate apparent clinical separation. This result should be presented as a strength rather than a weakness: the benchmark exposes the real generalization difficulty of GBM prognosis.

**The proposed model is promising as a mechanism, not yet as the top predictor.** The prior-guided kernel transformer recovered competitive 3-year performance but did not surpass the strongest classical baselines. The likely reasons are structural. First, the external branch currently omits methylation because TCGA probe harmonization is unfinished, so the model cannot yet exploit the modality that contributed most in the ablation study. Second, key clinical factors such as IDH mutation and MGMT promoter methylation are not yet integrated. Third, the scRNA prior bank is small. Recent glioma work combining EcoTyper-derived cell states and Self-Normalizing Networks reported much higher survival discrimination on larger mixed glioma cohorts [3]; our stricter GBM-only setting indicates that reproducing that level of performance requires both broader covariate coverage and more extensive cross-cohort harmonization.

**Practical next steps are now clear.** In the short term, the most valuable improvements are to harmonize TCGA methylation features for the external branch, add IDH/MGMT covariates, and report time-dependent survival metrics alongside fixed-horizon ROC. In the mid term, the scRNA branch should



## Single-cell GBM atlas across 17 CPTAC-3 cases



UMAP embeddings were taken from case-level Seurat outputs. Colors encode within-case Seurat clusters.

Fig. 4. Cohort-level single-cell atlas from 17 GBM cases. Each panel shows case-specific UMAP embedding colored by Seurat cluster assignment, with a summary strip reporting total cells and cluster richness per case.

move from recurrent marker transfer to richer state-level scores or EcoTyper-style ecosystem recovery. In the longer term, the prior-guided transformer should become a truly multimodal cross-attention model in which RNA, methylation, and transferred cell-state features are encoded in separate streams rather than concatenated after the fact.

## VI. CONCLUSION

We established a strict CPTAC→TCGA GBM benchmark that links bulk RNA, methylation, and scRNA-derived marker programs under a single reproducible workflow. The strongest current external models were the compact transformer for 1-year survival and LASSO logistic regression for 3-year survival, while the prior-guided kernel transformer remained

competitive only at the longer horizon. The ablation study showed that methylation is presently the dominant added modality, whereas the current six scRNA-derived programs contribute mainly interpretability rather than a large predictive gain. Taken together, the study contributes a realistic external benchmark, a biologically grounded prior-transfer mechanism, and a clear roadmap for improvement: harmonized methylation across cohorts, richer clinical covariates, and larger single-cell priors are needed before prior-guided multimodal transformers can claim clear predictive superiority in GBM survival modeling.

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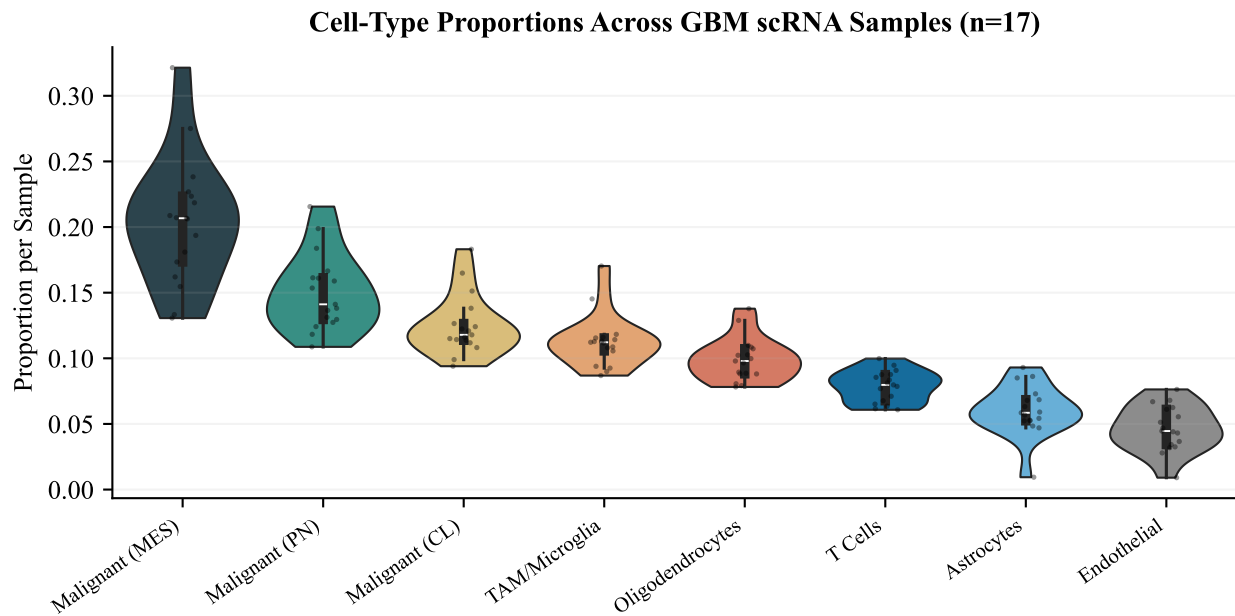


Fig. 5. Violin plot of cell-type proportions across 17 GBM scRNA-seq samples. Each point represents one patient sample. Proportions were derived from Seurat cluster assignments with putative cell-type annotation.

TABLE II  
STRICT CPTAC→TCGA EXTERNAL VALIDATION RESULTS.

| Model                                 | 1-Year AUC   | 1-Year C-index | 3-Year AUC   | 3-Year C-index |
|---------------------------------------|--------------|----------------|--------------|----------------|
| Logistic Regression                   | 0.548        | 0.540          | 0.592        | <b>0.586</b>   |
| LASSO Logistic                        | 0.550        | 0.533          | <b>0.612</b> | 0.558          |
| Ridge Regression                      | 0.474        | 0.494          | 0.565        | 0.563          |
| SVM (RBF)                             | 0.506        | 0.512          | 0.502        | 0.546          |
| Random Forest                         | 0.538        | 0.531          | 0.539        | 0.569          |
| Gradient Boosting                     | 0.519        | 0.519          | 0.546        | 0.542          |
| SNN                                   | 0.544        | 0.529          | 0.504        | 0.526          |
| Compact Transformer                   | <b>0.581</b> | <b>0.553</b>   | 0.599        | 0.544          |
| Prior-Guided Kernel Transformer + SVM | 0.516        | 0.498          | 0.586        | 0.542          |

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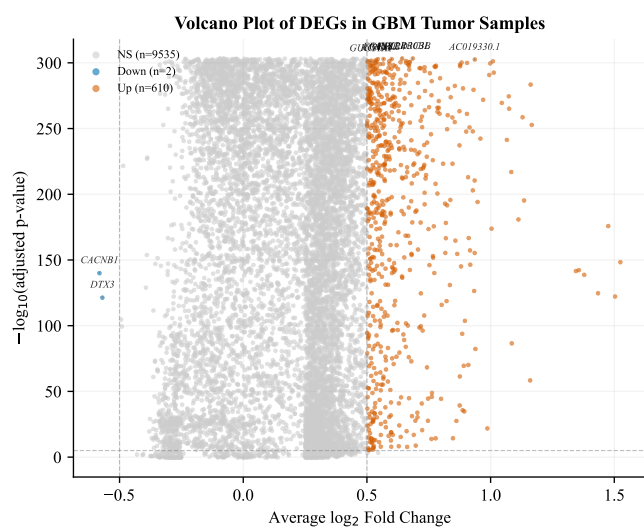


Fig. 6. Volcano plot of DEGs from GBM scRNA-seq data. Upregulated (red,  $n = 610$ ) and downregulated (blue) genes are highlighted. Top genes are labeled. Thresholds:  $|\log_2\text{FC}| > 0.5$ ,  $-\log_{10}(\text{adj-}p) > 5$ .





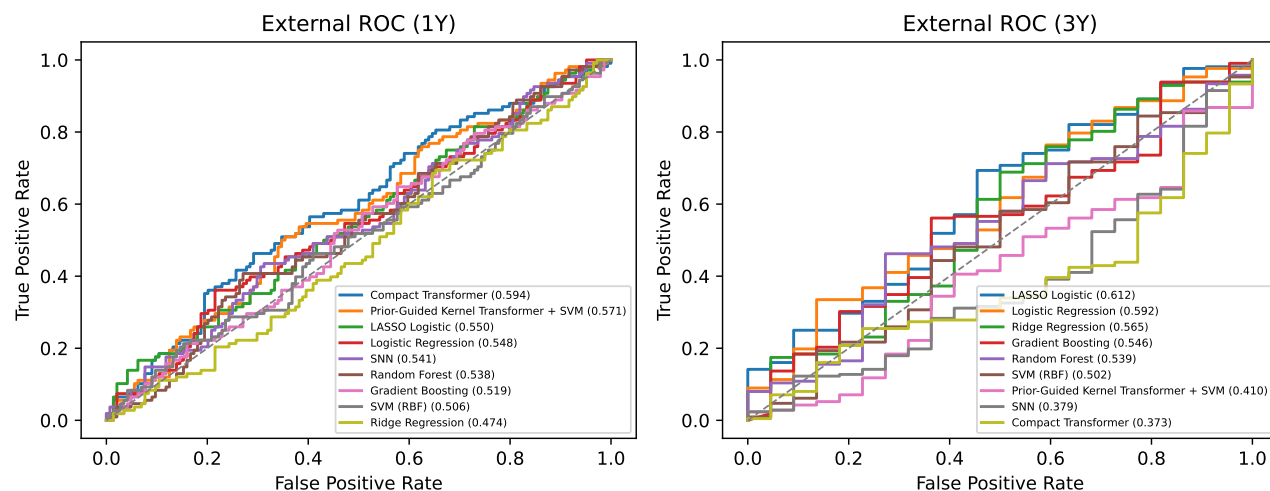


Fig. 9. External ROC curves on TCGA for all v7 models. Left: 1-year endpoint. Right: 3-year endpoint. The compact transformer leads at 1 year, while LASSO logistic leads at 3 years.

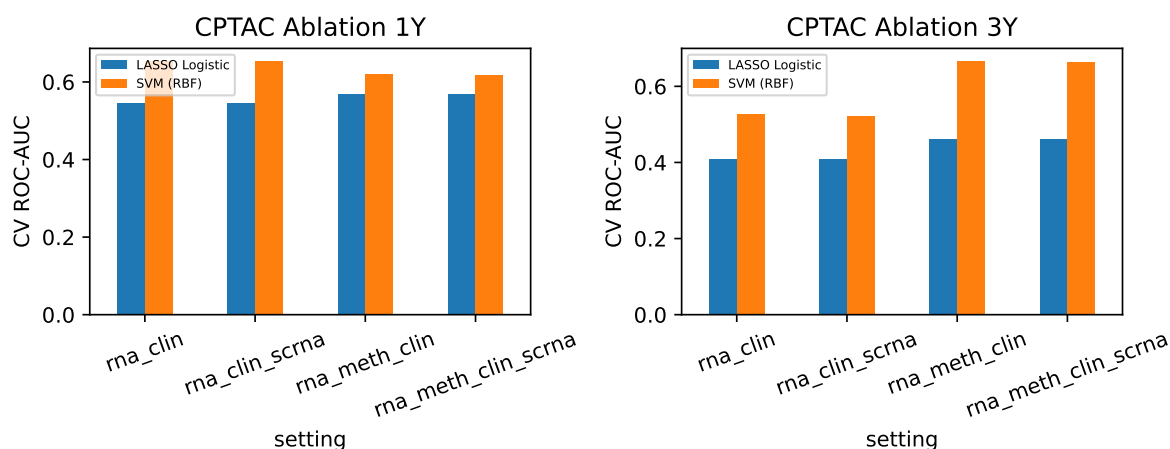


Fig. 10. CPTAC internal ablation across four feature settings. Methylation provides the strongest incremental signal, especially for 3-year survival, whereas the current six scRNA-derived program scores induce only small changes.

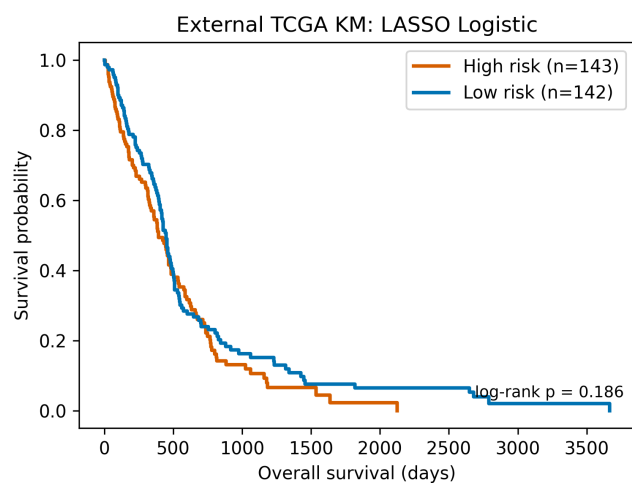


Fig. 11. External TCGA Kaplan–Meier curves using the best 3-year external model (LASSO logistic). Median-split risk groups show a survival trend, but the separation is not yet statistically significant.